

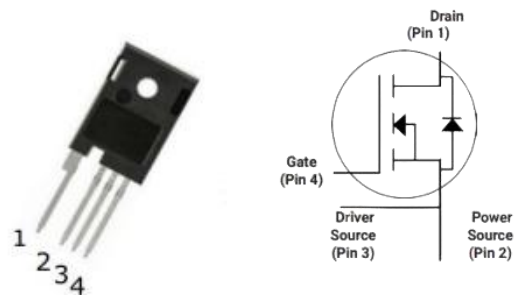
SiC N-Channel MOSFET

FEATURES

- Optimized Package with Separaye Driver Source Pin
- High Bloking Voltage with Low On-Resistance
- High Speed Switching with Low Capability

APPLICATIONS

- Capacitor Discharge
- Solar Inverters
- Switch Mode Power Supplies
- High Voltage DC-DC Converters



Absolute Maximum Ratings(TC=25°C)

SYMBOL	PARAMETER	VALUE	UNIT
V _{DSS}	Drain-Source Voltage	1700	V
V _{GS(max)}	Gate-Source Voltage	-10/+25	V
V _{GSop}	Gate-Source Voltage	-5/+20	V
I _D	Drain Current-Continuous@T _c =25°C	40	A
	Drain Current-Continuous@T _c =100°C	27	A
I _{DM}	Drain Current-Single Pluse	80	A
P _D	Total Dissipation @T _c =25°C	277	W
T _J	Max. Operating Junction Temperature	150	°C
T _{stg}	Storage Temperature	-55~150	°C

THERMAL CHARACTERISTICS

SYMBOL	PARAMETER	MAX	UNIT
R _{th j-c}	Thermal Resistance, Junction to Case	0.45	°C/W

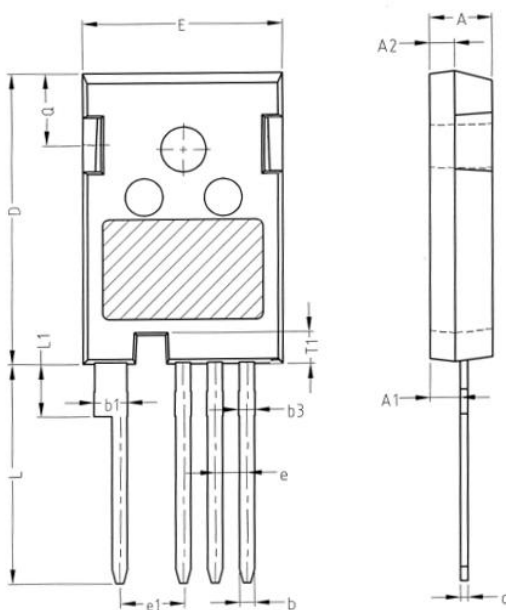
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ELECTRICAL CHARACTERISTICS (T_c=25°C)

SYMBOL	PARAMETER	CONDITIONS	MIN	TYP.	MAX	UNIT
V _{(BR)DSS}	Drain-Source Breakdown Voltage	V _{GS} = 0V, I _D = 100uA	1700	--	--	V
I _{GSS}	Gate Source Leakage Current	V _{GS} = -5/+20V, V _{DS} = 0V	--	--	±250	nA
I _{DSS}	Zero Gate Voltage Drain Current	V _{DS} = 1700V, V _{GS} = 0V	--	--	100	uA
V _{GS(th)}	Gate Threshold Voltage	V _{DS} = V _{GS} , I _D = 10mA	2.0	--	4.0	V
R _{DS(ON)}	Drain-Source On-stage Resistance	V _{GS} = 20V, I _D = 28A	--	--	125	mΩ
V _{SD}	Diode Forward Voltage	I _{SD} = 10A, V _{GS} = -5V	--	4.2	--	V

PACKAGE OUTLINE: TO-247-4

COMMON DIMENSIONS



SYMBOL	MM		
	MIN	NOM	MAX
A	4.80	5.00	5.20
A1	2.21	2.41	2.61
A2	1.80	2.00	2.20
b	1.06	1.21	1.36
b1	2.33	2.63	2.93
b3	1.07	1.30	1.60
c	0.51	0.61	0.75
D	23.30	23.45	23.60
D1	16.25	16.55	16.85
E	15.74	15.94	16.14
E1	13.72	14.02	14.32
T1	2.35	2.50	2.65
e	2.54 BSC		
e1	5.08 BSC		
Q	5.49	5.79	6.09
L	17.27	17.57	17.87
L1	3.99	4.19	4.39
Φ _p	3.40	3.60	3.80
Φ _{p1}	7.19 REF		



ISCF8119

eq C2M0080170P

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